Homework 4

Referring Expression Segmentation (RES)

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Introduction

- In this assignment, you will fine-tune a **language-guided image segmentation model** on the given dataset to learn how natural-language inputs can condition pixel-level predictions.

What to do?

- You’re tasked with completing at least 4 Python Files: ***dataset.py, model.py, train.py, test.py***
- You need to implement your model inside ***model.py***. This network should be rendered accessible to ***train.py*** and ***test.py*** through import.
- Within these files, you can import any package and design any additional classes or functions if you need.
- **Discovery of any infringement of this cardinal rule will incur a penalty of a zero score for this assignment.**

What to do?

- We have already implemented portions of the provided template. However, if you prefer, you may discard our implementation entirely.
- You can ONLY use **openai/clip-vit-base-patch16** as your base model.

```
def __init__(self, image_size=IMAGESIZE):
    super().__init__()
    # You Shall NOT modify the loaded model!
    # Load CLIP (ViT-B/16)
    self.clip = CLIPModel.from_pretrained("openai/clip-vit-base-patch16")
    self.image_size = image_size
```

Dataset

The dataset is built upon COCO images and contains a total of 17,596 samples for evaluation:

- Train: 13,811 samples
- Valid: 1,975 samples
- Test: 1,810 samples

Grading (Performance 80% + Ranking 20%)

- ***oIoU (80 points)***

  40% <= oIoU, get full points

  37% <= oIoU < 40%, get 70 points

  35% <= oIoU < 37%, get 60 points

  32% <= oIoU < 35%, get 50 points

  oIoU < 32%, get 0 points

- ***oIoU Ranking (20 points)***

  This evaluation is based on the oIoU of your model.

  The higher oIoU you achieve, the higher score you will get.

  Score formula: $\text{Round}\left(\frac{n-r}{n-1} \times 20\right)$

  n: number of students

  r: your ranking (r = 1 being the best rank)

Grading

- Your ***test.py*** need to generate **"pred_{student_id}.csv"** based on the weight(.pth) you provided.
- Make sure your output format match the format we want.

submission.csv

```
id,image,segmentation
12345,012345.jpg,"[398.0, 6.0, 397.0, 47.0, 395.0, 47.0, 394.0, 48.0, 393
23456,023456.jpg,"[160.0, 6.0, 159.0, 377.0, 155.0, 377.0, 154.0, 378.0,
34567,034567.jpg,"[146.0, 3.0, 145.0, 24.0, 143.0, 24.0, 141.0, 26.0, 141
45678,045678.jpg,"[179.0, 5.0, 178.0, 59.0, 175.0, 59.0, 173.0, 61.0, 172
56789,056789.jpg,"[294.0, 12.0, 293.0, 68.0, 289.0, 68.0, 288.0, 69.0, 28
```

Your pred_{student_id}.csv should looks like.

Submission

- Your submission should contain:

  - Model Structure: model.py

  - Data handler: dataset.py

  - Train: train.py

  - Test: test.py

  - [other python files you add]

  - Model Weight: w_{student_id}.pth

  - Prediction: pred_{student_id}.csv

```
hw4_{student_id}.zip
├ model.py
├ dataset.py
├ train.py
├ test.py
├ [other python files you add]
├ w_{student_id}.pth
└ pred_{student_id}.csv
```

Submission

- Compress all files into a single zip file named hw4_{student_id}.zip without any extra folder layers.
- Make sure not to include unnecessary files or folders (e.g., .DS_Store, __MACOSX). The submission must not contain the dataset.

Penalty

- Format penalty - **20 points**

  - If you have any incorrect file format or name, then you will get **-20 points.**

- Late penalty - **10% per day**
- **Deadline: 2025/12/26 23:59**

Note

- If you have any questions, please contact us via the E3 system and include **ALL** TAs in your message